SIGHTLINE SCHEMATIC · STEP-BY-STEP GUIDE 3 OF 3

Revit: create the bowl profile families

Download the Revit package and run the Dynamo graph to create one Revit profile family per tier, ready for a sweep.

YOU NEED

- Revit 2024 with Dynamo 2.17
- A Revit project open (any project, even a new one)
- The bowl you want, open in Sightline Schematic

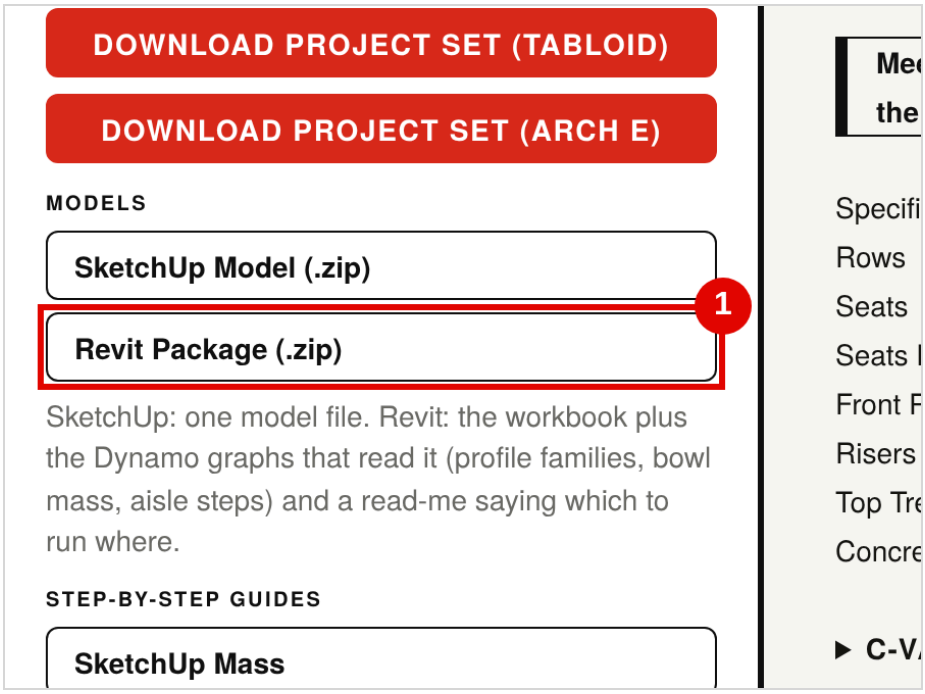
YOU GET

- One Revit profile family (.rfa) per tier, saved in a folder you pick and loaded into the project
- Each profile placed at its true distance from the point of focus, so every tier sweeps along the same path

STEPS

1 Download the Revit package

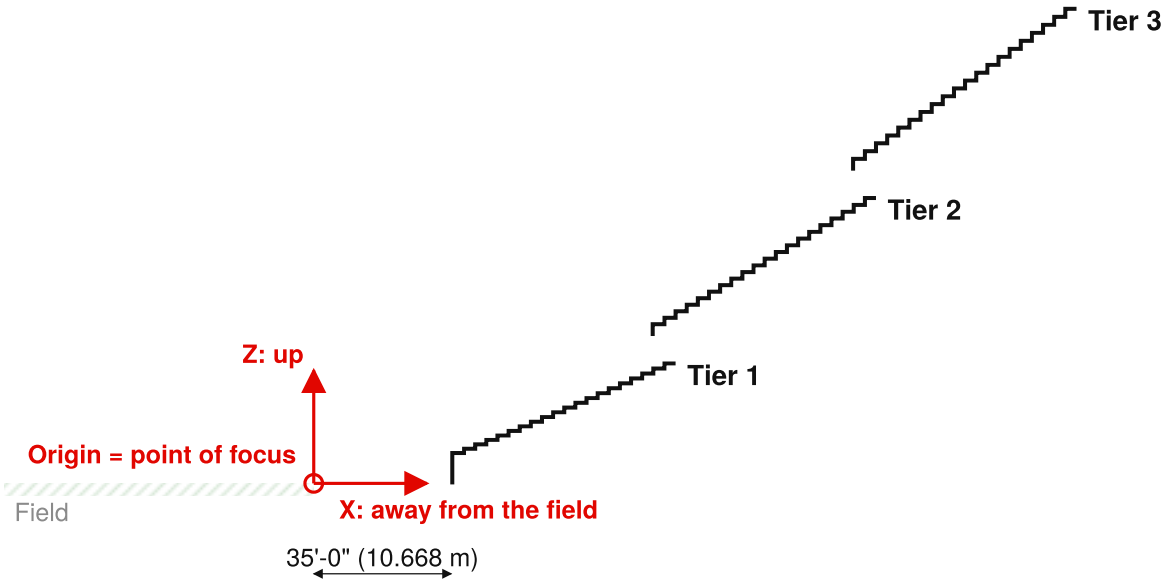
In Sightline Schematic, under **Deliver › Models**, click **Revit package (.zip)** and unzip **Stadium_Bowl_Revit_R2024.zip** to a folder you keep.



Deliver › Models. (1) Revit package (.zip).

Stadium_Bowl_Revit_R2024.zip (the two files this guide uses)	
XLSX **Stadium_Bowl_Excel_YYYYMMDD_hhmmss.xlsx**	bowl data, input 1 of every graph
DYN **Stadium_Bowl_Profile_R2024.dyn**	profile families

2 Know what the profile is



The tier profiles of this bowl, drawn from the workbook. Each family holds one tier as a closed loop (treads, risers, slab underside and leg wall). Its origin is the point of focus: the field edge at pitch level.

Because every family shares the point of focus as its origin, Tier 2 and Tier 3 already sit at their true distance and height (86'-0" and 137'-0" to the front riser in this example). You never move a profile by hand.

3 Open a project and start Dynamo

Manage › **Visual Programming** › **Dynamo**

Open the project the bowl belongs in (or a new project). In Dynamo, **File › Open** **Stadium_Bowl_Profile_R2024.dyn**, run mode **Manual**. Dynamo Player works too.

4 Set the five inputs and run

Stadium_Bowl_Profile_R2024.dyn		Inputs
1	Sightline workbook (.xlsx)	Stadium_Bowl_Excel_YYYYMMDD_hhmmss.xlsx
2	Profile family template (.rft)	C:\ProgramData\Autodesk\RVT 2024\Family Templates\English_I\Profile.rft
3	Save the families in	…\Stadium\Revit\Families\
4	Family name prefix	Stadium_Bowl_Profile
5	Load into the open project	True
		Manual **Run**

Schematic of the graph inputs. Input 2 already points at the standard imperial template.

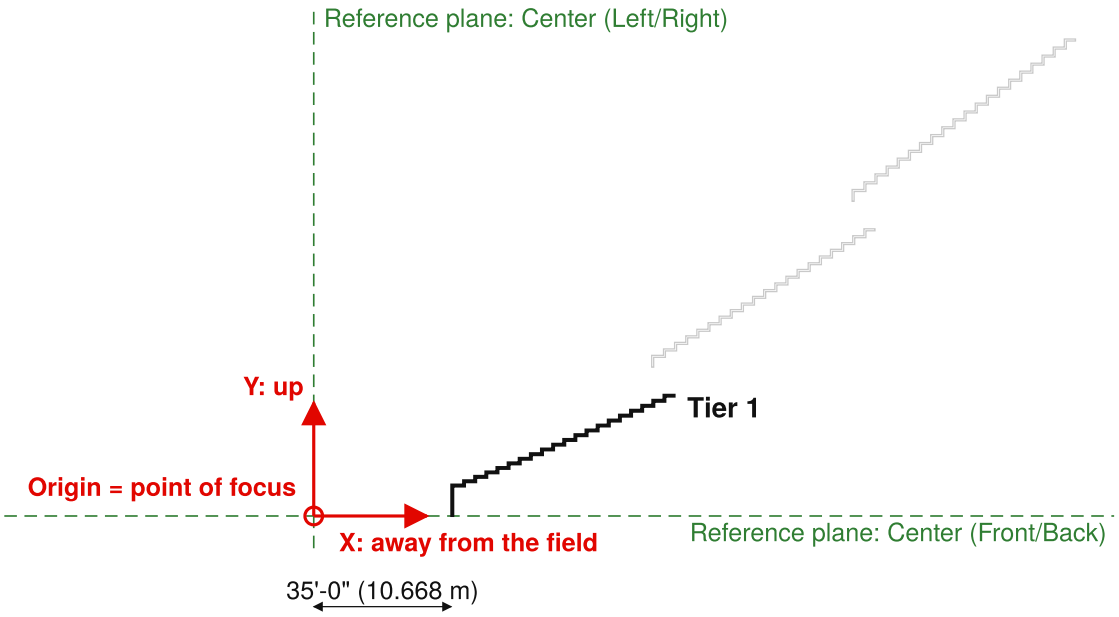
- **Input 2**: metric offices use `…\English\Metric Profile.rft`. If your templates live elsewhere, see **File › Options › File Locations › Default path for family template files**.
- **Input 4** names the families: `Stadium_Bowl_Profile_T1`, `_T2`, `_T3`. When a tier's sides differ (for example a deeper East sideline), you get one per side: `Stadium_Bowl_Profile_T1_E`, `_T1_SEZ` and so on.
- Click **Run**. The **Result** node lists each saved .rfa, its line count and “loaded”. Running again overwrites and reloads families of the same name.

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5 Check the families

Project Browser › **Families** › **Profiles** › **Stadium_Bowl_Profile_T1**

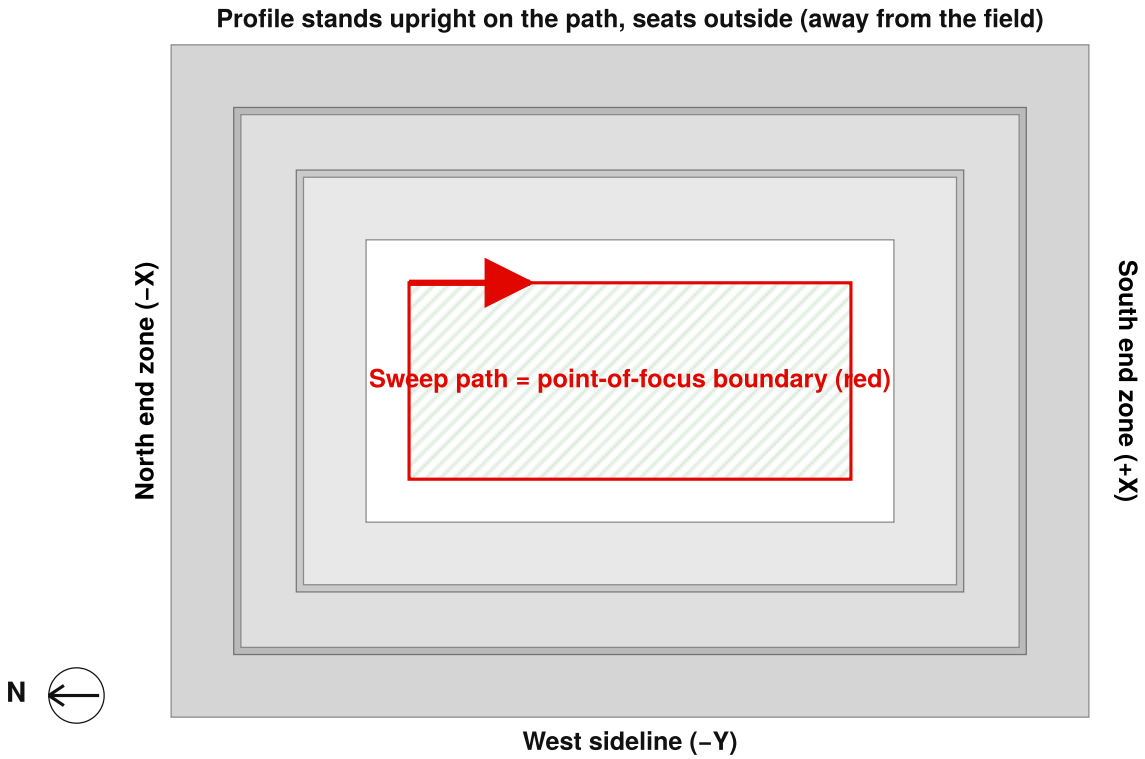
To look inside, right-click the family and choose **Edit**. The loop sits on the Ref. Level plan, with the point of focus where the two Center reference planes cross:



Inside Stadium_Bowl_Profile_T1.rfa: one closed loop of lines. Family X runs away from the field, family Y is up (the workbook's Z). The grey tiers are for reference only; they are in their own families.

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6 Draw the sweep path: the point-of-focus boundary



The sweep path is the point-of-focus boundary: the field edge Sightline sightlines are measured to. Every tier sweeps along this same path.

Architecture › **Component** › **Model In-Place**

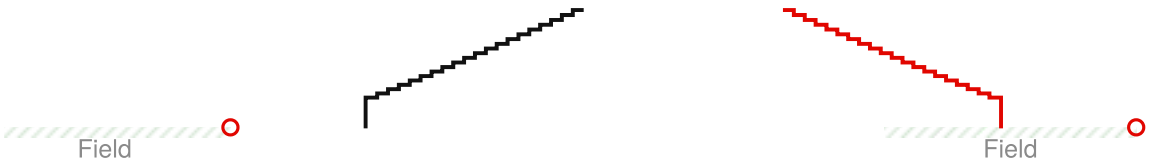
- Category **Generic Models** (or Mass), name it, for example **Bowl Tier 1**.
- **Create › Forms › Sweep**, then **Sketch Path**. Draw the boundary with the Rectangle tool from the field corners. Its size is the Field line on the workbook Summary sheet (109.73 × 48.77 m here); the corners are on the **Focal path** sheet. Click the green check to finish the path.

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7 Pick the profile and check its direction

- In the **Sweep** panel, set **Profile** to **Stadium_Bowl_Profile_T1**. Revit places the profile at the red path point.
- Look at a 3D or section view. The seats must rise away from the field. If they lean over the field, tick **Profile is Flipped** in Properties (or set Angle to 180°).

Section through the sweep at the path (red circle = path)



Correct: seats rise away from the field **Flipped: seats over the field. Flip the profile**

Left: correct. Right: flipped, the tier hangs over the field. The path direction decides which way Revit first places the profile.

Click the green check twice (finish the sweep, then **Finish Model**).

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8 Repeat for each tier

- Make one sweep per tier, all on the same path, with `_T2` and `_T3`. They land in the right place on their own.
- Families per side (`_T1_E` and so on): sweep each one along its own side of the boundary (Pick Path or a single line).
- A sweep mitres the corners. Sightline's chamfered or open corners are built exactly by the bowl mass graph (guide 2); use the profiles for sections, sweeps of one stand, slab edges and details.

IF THE GRAPH STOPS

Message in the Result node	What to do

2. Pick the profile family template	The template path is wrong. Browse to Profile.rft (imperial) or Metric Profile.rft (metric).
3. Pick a folder to save the profile families in	Input 3 is empty or the folder no longer exists.
No "T1 E profile" sheets in this workbook	The workbook is not from the Revit package. Download the package again.
Families saved but not loaded	A family document was active. Open a project, make it active and run again (Load True).
The profile is not in the Sweep's list	Load it with **Insert › Load Family** from the folder of input 3.

Sightline Schematic 2026.11.05.03. Guide file: Stadium_Bowl_Profile_R2024_Guide.pdf. Menu names are for SketchUp 2024 and Revit 2024 (Dynamo 2.17); other versions may differ slightly.